

VDM

The Incubator case study

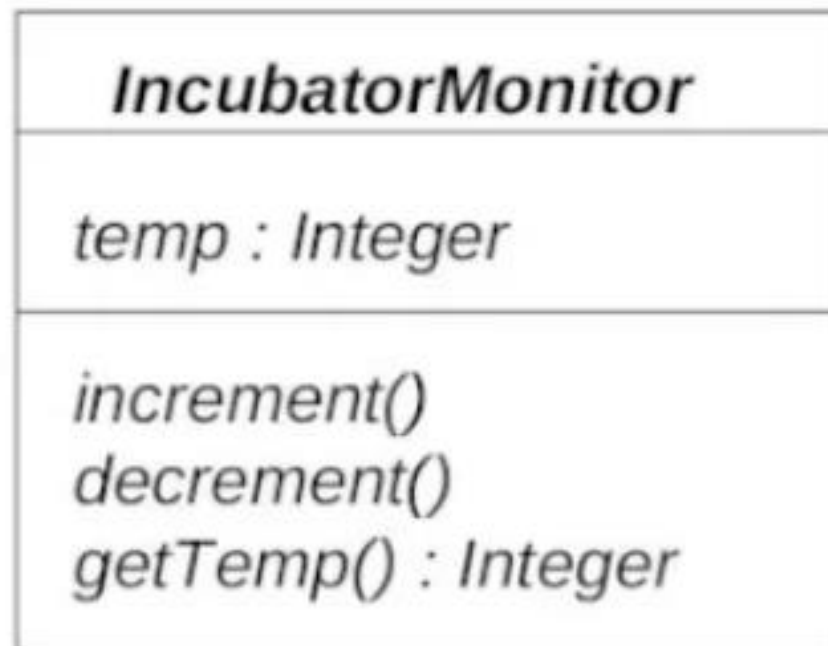
The temperature of the incubator needs to be carefully controlled and monitored;

Safety requirements :



-10 Celsius ← TEMPERATURE → +10 Celsius

The UML specification



Specifying the 'state' in VDM-SL

The VDM state refers to the permanent data stored by the system.

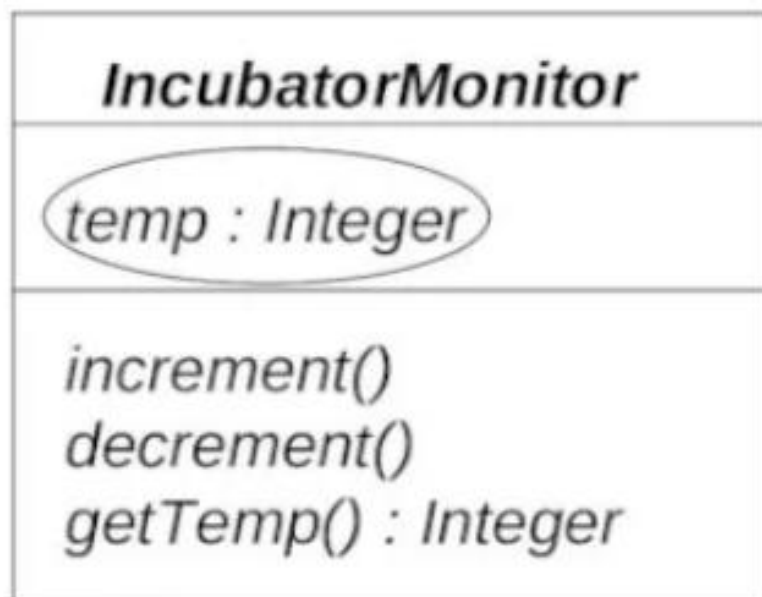
<i>IncubatorMonitor</i>
<i>temp : Integer</i>
<i>increment()</i> <i>decrement()</i> <i>getTemp() : Integer</i>

In VDM-SL we use **mathematical** types

TYPES AVAILABLE IN VDM

- $\mathbb{N}$: natural numbers (positive whole numbers)
- $\mathbb{N}_1$: natural numbers excluding zero
- $\mathbb{Z}$: integers (positive and negative whole numbers)
- $\mathbb{R}$: real numbers (positive and negative numbers that can include a fractional part)
- $\mathbb{B}$: boolean values (true or false)
- Char* : the set of alphanumeric characters

UML



VDM-SL

state *IncubatorMonitor* **of**

temp : $\mathbb{Z}$

end

Operations in VDM SL

<i>IncubatorMonitor</i>
<i>temp : Integer</i>
<i>increment()</i> <i>decrement()</i> <i>getTemp() : Integer</i>



Each operation specified in VDM-SL as follows:

the operation header
the external clause
the precondition
the postcondition

Increment Operation

<i>IncubatorMonitor</i>
<i>temp : Integer</i>
<i>increment()</i> <i>decrement()</i> <i>getTemp() : Integer</i>

increment()

ext *wr temp : □*

pre *temp < 10*

post $\overline{temp} > \overline{temp}$

$\overline{temp} + 1 = temp$

Decrement Operation

<i>IncubatorMonitor</i>
<i>temp : Integer</i>
<i>increment()</i> <i>decrement()</i> <i>getTemp() : Integer</i>

decrement()

ext **wr** *temp* : $\square$

pre *temp* > -10

post *temp* = $\overline{\text{temp}}$ - 1

Get Temperature

<i>IncubatorMonitor</i>
<i>temp : Integer</i>
<i>increment() decrement() getTemp() : Integer</i>

```
getTemp( ) currentTemp : []  
ext rd temp : []  
pre TRUE  
post currentTemp = temp
```